

$$\begin{array}{c}
 \mathbf{A}' \\
 \begin{array}{c} i_h o_h o_w \\ \left[\begin{array}{|c|c|c|c|} \hline a_{00} & a_{01} & a_{02} & a_{03} \\ \hline a_{10} & a_{11} & a_{12} & a_{13} \\ \hline a_{20} & a_{21} & a_{22} & a_{23} \\ \hline a_{30} & a_{31} & a_{32} & a_{33} \\ \hline \end{array} \right] \end{array}
 \end{array}
 \begin{array}{c}
 * \\
 \begin{array}{c} k_h k_w \\ \left[\begin{array}{|c|} \hline b_{00} \\ \hline b_{01} \\ \hline b_{10} \\ \hline b_{11} \\ \hline \end{array} \right] \end{array}
 \end{array}
 \begin{array}{c}
 \mathbf{B}' \\
 \begin{array}{c} k_c \\ \left[\begin{array}{|c|} \hline b_{00} \\ \hline b_{01} \\ \hline b_{10} \\ \hline b_{11} \\ \hline \end{array} \right] \end{array}
 \end{array}
 =
 \begin{array}{c}
 \mathbf{C}' \\
 \begin{array}{c} o_h \\ \left[\begin{array}{|c|} \hline c_{00} = a_{00} \times b_{00} + a_{01} \times b_{01} + a_{10} \times b_{10} + a_{11} \times b_{11} \\ \hline c_{01} = a_{02} \times b_{00} + a_{03} \times b_{01} + a_{12} \times b_{10} + a_{13} \times b_{11} \\ \hline c_{10} = a_{20} \times b_{00} + a_{21} \times b_{01} + a_{30} \times b_{10} + a_{31} \times b_{11} \\ \hline c_{11} = a_{22} \times b_{00} + a_{23} \times b_{01} + a_{32} \times b_{10} + a_{33} \times b_{11} \\ \hline \end{array} \right] \end{array}
 \end{array}
 \begin{array}{c}
 o_w \\
 \left[\begin{array}{|c|} \hline c_{00} = a_{00} \times b_{00} + a_{01} \times b_{01} + a_{10} \times b_{10} + a_{11} \times b_{11} \\ \hline c_{01} = a_{02} \times b_{00} + a_{03} \times b_{01} + a_{12} \times b_{10} + a_{13} \times b_{11} \\ \hline c_{10} = a_{20} \times b_{00} + a_{21} \times b_{01} + a_{30} \times b_{10} + a_{31} \times b_{11} \\ \hline c_{11} = a_{22} \times b_{00} + a_{23} \times b_{01} + a_{32} \times b_{10} + a_{33} \times b_{11} \\ \hline \end{array} \right]
 \end{array}
 \end{array}$$

Diagram illustrating the matrix multiplication of $\mathbf{A}'$ and $\mathbf{B}'$ to produce $\mathbf{C}'$.

Matrix $\mathbf{A}'$: Dimensions $i_h o_h o_w$ by $k_h k_w k_c$. The matrix is partitioned into four quadrants: top-left (yellow), top-right (orange), bottom-left (cyan), and bottom-right (blue).

Matrix $\mathbf{B}'$: Dimensions $k_h k_w$ by k_c . The matrix is a column vector partitioned into two segments: top (white) and bottom (white).

Matrix $\mathbf{C}'$: Dimensions o_h by o_w . The matrix is a row vector partitioned into four segments: top (yellow), second (orange), third (gray), and bottom (blue).

The multiplication is shown as $\mathbf{A}' * \mathbf{B}' = \mathbf{C}'$.

The resulting matrix $\mathbf{C}'$ contains the following elements:

- $c_{00} = a_{00} \times b_{00} + a_{01} \times b_{01} + a_{10} \times b_{10} + a_{11} \times b_{11}$
- $c_{01} = a_{02} \times b_{00} + a_{03} \times b_{01} + a_{12} \times b_{10} + a_{13} \times b_{11}$
- $c_{10} = a_{20} \times b_{00} + a_{21} \times b_{01} + a_{30} \times b_{10} + a_{31} \times b_{11}$
- $c_{11} = a_{22} \times b_{00} + a_{23} \times b_{01} + a_{32} \times b_{10} + a_{33} \times b_{11}$